# The Compiler Closure of TFPT

What concretely simplifies, reduces and closes relative to the seven original papers — a status assessment and reading guide

June 12, 2026

**In one sentence**

The theory tips from "many sectors with surprising hits" (the state of Papers 1–7) to "*one small discrete compiler generates those sectors*": $E_8$ — which in v4.5 appeared only *peripherally* (the orbit cascade for scale ordering) — is now *built* from two TFPT atoms and moved to the centre; the flavor matrix, the fine-structure constant and the scale grammar follow as *fixed points* from exactly two inputs $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$.

**The TFPT document set — what is treated where**

**Plain language:** TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus Appendix H**, best read in order (this is the entry point; a detailed map is in §1):

1. `introduction` — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point $\alpha^{-1}$, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. `tfpt_2_standard_model` — the SM in one $\varphi_0$-ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. `tfpt_3_e8_audit_bootstrap` — the seven $E_8$ slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. `tfpt_4_frontier` — honest status of $\eta_B$, $m_p/m_e$, Koide, dark matter, full QG.

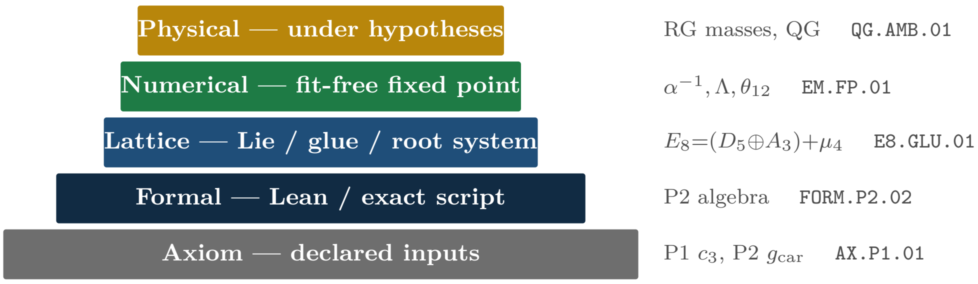
H. `tfpt_horizon_readouts` — **Appendix H** (unit-system reframe, *not* new physics): one seam constant as the universal horizon thermal code.

O. `origin_theory` — **Origin Theory**: why no free fundamental number remains. The $(g_{\text{car}}, N_{\text{fam}})$=(5, 3) skeleton, the triply-forced 8 (geometry = lattice = gravity), the order-30 Coxeter cycle, one boundary transport for flavor and horizon, and the gapped *unique attractor*. [I]/[L] core (`v53–v56`) plus one honestly-typed [P] cyclic interpretation.

**You are reading document #1.**

## Contents

**Reading the markers (refined).** The former overloaded "[M]" is split into the grade it actually carries, so a reviewer can see at a glance which claims are exact, which are formal, and which are conditional: [I] *exact identity* (e.g. $240 = 16 \cdot 5 \cdot 3$); [L] *Lie/lattice theorem* (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$); [F] *formalised* (Lean- or script-verified); [N] *numerical fixed point* (reproduced, stable); [P] *physical/conditional* (follows under named hypotheses); [A] *axiom or open*.




*The claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv` (Fig. 1).

**No-free-pattern rule.** To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [I] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

**Admissible-invariant classes (the harder filter).** Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of $(R, Q, K, L)$; (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a \rangle$; (vi) an $E_8$ Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

**Reviewer path (read in this order; the rest is support)**

1. the architecture: the two axioms $\{c_3, g_{\text{car}}\}$ and the two-engine picture (§1, the figures);
2. the $E_8$ glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked

(verification/v1_e8_glue.py);

3. the $\alpha^{-1}$ fixed point plus ablation (v3_em_alpha.py);
4. the **single status ledger** verification/status_ledger.csv — the source of truth for every claim's grade, location, dependency and script;
5. the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

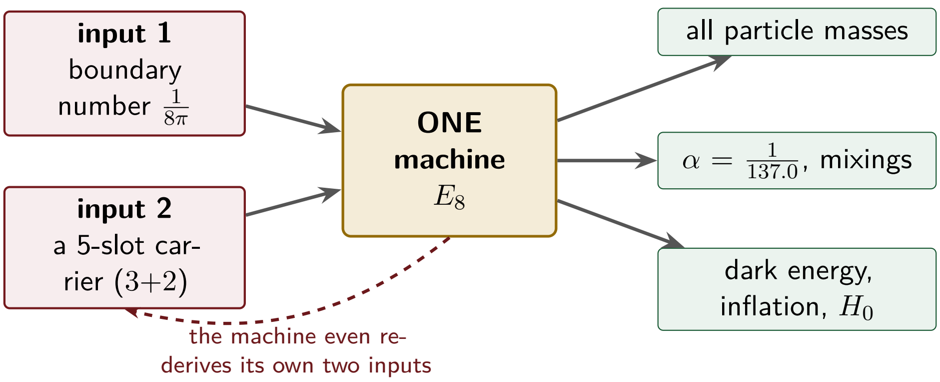
# 1 What this is about: the jump in construction principle

**In plain words — what we found, and why it matters**

**What we found.** The older version of the theory already reproduced dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs. What it could *not* say was *why* the same handful of small integers $(2, 3, 5, 16, \dots)$ kept reappearing in every sector. The new result closes exactly that gap: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry $E_8$, and $E_8$ is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

**How this drastically simplifies the theory.** Instead of seven separate derivations that happen to share numbers, there is now *one generator and a handful of corollaries*. Roughly 250 pages of sector-by-sector work collapse to "two inputs + one machine"; the number of independent structural assumptions drops to **two**.

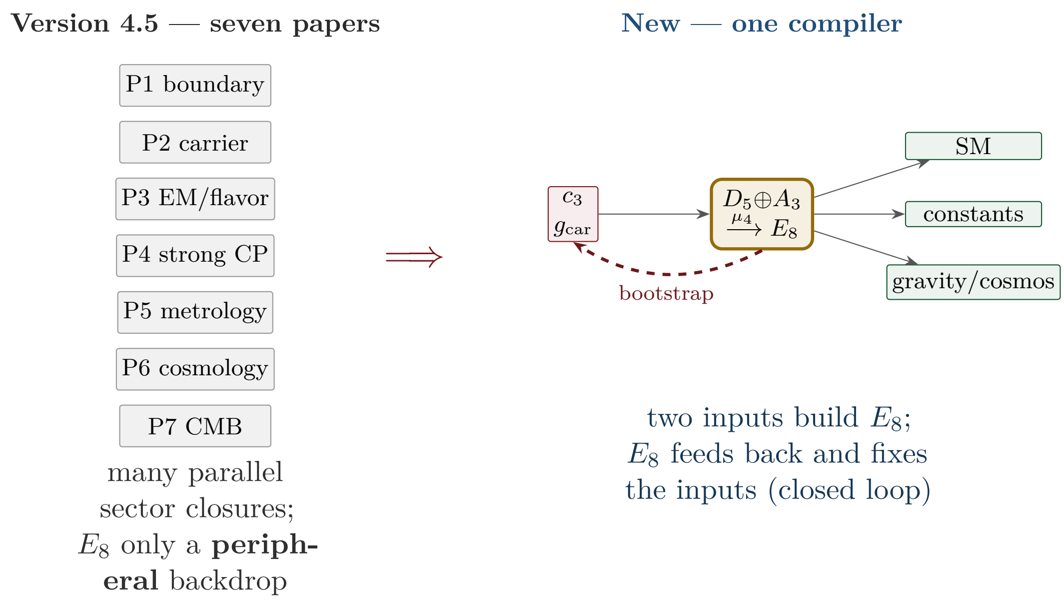
**How plausible it is.** A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed loop below), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number $\pi$.




*Two numbers go in on the left; the discrete Standard-Model core, the dimensionless constants and several scale readouts come out on the right — and the dashed loop is the point: the machine reproduces the very two numbers it started from, so the discrete core is* overdetermined *rather than fitted. (Dimensionful EW/QCD masses, the full quantum-gravity measure and the frontier items stay typed-open — see the status ledger.)*

The seven original papers (tfpt-45, Papers 1–7) deliver a long chain of physical closures: boundary kernel → carrier → Standard-Model packet → EM fixed point → flavor transport →

admissibility/strong CP → geometric gravity/metrology → cosmology/CMB. Each stage stands on its own and hits a remarkable number of values. What was missing was a *meta-statement* explaining *why* so many independent sectors use the same small stock of numbers $(2, 3, 5, 16, 40, 41, 48, 240, 248)$.




**The difference in one picture.** v4.5 is a *stack of seven independent closures* that happen to share numbers, with $E_8$ assumed in the background. The new approach is *one discrete compiler*: two inputs build $E_8$, $E_8$ outputs the sectors, *and* $E_8$ feeds back to fix the two inputs (the red loop). What were seven parallel results become one generator with corollaries; what was a posited $E_8$ becomes a constructed-and-self-checking closure.

**Sharper still: one anchor $a = (1, 1, 2)$, and $E_8$ as a *compiler hull* [I]/**

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\rm car}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n+1^n+2^n = 2+2^n$ generate the big Lie data directly: $p_1=4=|\mu_4|$, $p_2=6=|R^+(A_3)|$, $p_3=10=\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1p_2p_3 = 240, \quad \dim E_8 = p_1p_2p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\rm car}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$. [`verification/v23_anchor_generator.py`]

*And $a$ is itself half-forced* ([L]/[P]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\rm fam} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2+1+1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not "why $a = (1, 1, 2)$" but "why the carrier $g_{\rm car} = 5$ / $(|\mu_4|, N_{\rm fam}) = (4, 3)$" — that interface (with $c_3$) is the [A] input layer.

**What $E_8$ is (and is not).** In TFPT $E_8$ is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct $E_8$ gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal $E_8$ world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain $E_8$ as a spacetime gauge symmetry, which

TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the $E_8$ audit hull.*

In one line:

$$\boxed{\text{The anchor generates the syntax; the boundary generates the scale; } E_8 \text{ checks the consistency.}}$$

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators $R, Q, K, L$) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, $\varphi_0$, $\alpha^{-1}$, the exponential scales) flows from $\pi$; and $E_8$ is the unimodular *checksum container*, not the world group.

**The anchor ratio triad (** [`verification/v119_review_validation_2.py`]**).** The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions [I]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio)}, \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain)}, \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [I]: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2 \operatorname{rank} E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\operatorname{rank} E_8 = p_0 + e_2$.

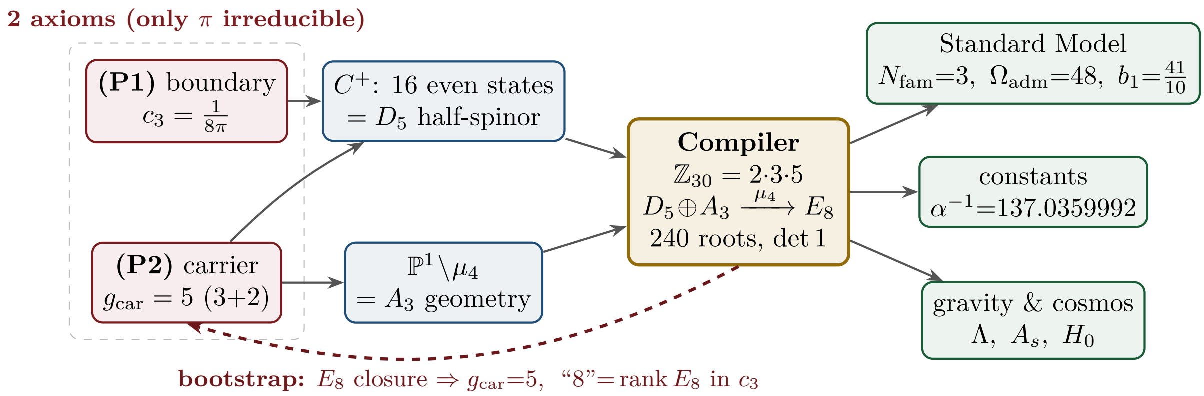
That is now the case. The whole development is consolidated into **five companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the earlier notes of its layer into self-contained parts:

| **Document** | **What it contains (former notes now as parts)** |
|---|---|
| `tfpt_1_architecture_e8` | **Core.** The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}$=2·3·5, the explicit $D_5 \oplus A_3 + \mu_4$ $E_8$ construction, the machine-checkable $E_8$ appendix and the group-level glue $(\text{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ (Part II). |
| `tfpt_2_standard_model` | **Standard Model.** The full SM in one $\varphi_0^{\text{ret}}$-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/$n_s$, $r$, Starobinsky $M$, derived $\theta_{12}$, quark $c$, the H2 splitting type, Λ-typing (Part III). |
| `tfpt_3_e8_audit_bootstrap` | **$E_8$ audit & bootstrap.** The seven $E_8$ slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade $D$=60−2$n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}$=5 and the "8" in $c_3$ as overdetermined fixed points, only $\pi$ irreducible (Part III). |
| `tfpt_4_frontier` | **Frontier.** The honest status of $\eta_B$, $m_p/m_e$, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder. |
| `tfpt_horizon_readouts` **(App. H)** | **Appendix H — horizon unit system (reframe, not new physics).** One seam constant $c_3$=$\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}$=$c$ — standard physics in seam units, two compiler fingerprints (1920=$\lvert W(D_5)\rvert$, $\lvert\mu_4\rvert$=4). |

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the $\theta_{12}$ texture and the quark $c$ in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a $U_{\text{point}}$ anchor; $\theta_{12}$ the seam-misalignment residual). Five companion files (plus this introduction) instead of a dozen, each result where it logically lives.

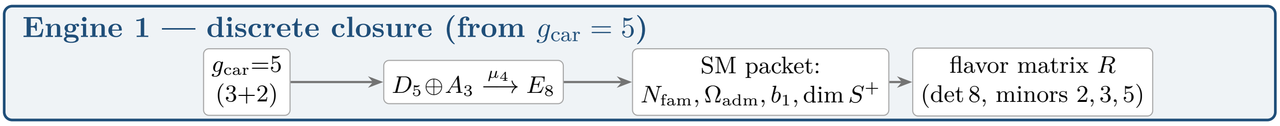
## 2 The architecture at a glance

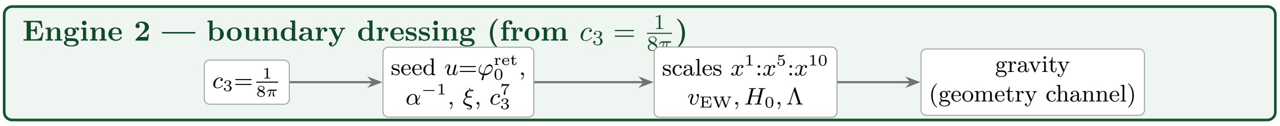
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.




**Key statement (the loop closes):** previously $D_5$, $A_3$ and $E_8$ sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the $E_8$ closure feeds back and *fixes the inputs*. $g_{\mathrm{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0}+\binom{g}{1}+\binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output are mutually locked — they prove each other**; the only thing *not* produced by the loop is the continuous $\pi$ (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed self-consistency loop.

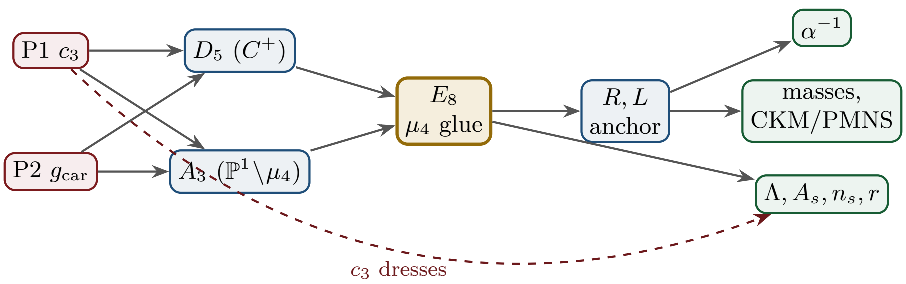
**The master story is two engines.** Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.




## 3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



| Node | Definition | Inputs | St. | Output | Failure mode |
|---|---|---|---|---|---|
| P1 | RP boundary kernel, $c_3$=1/(8$\pi$) | — (axiom) | [A] | $c_3$ | no reflection-positive seam |
| P2 | 5-slot carrier $g_{\text{car}}$=5 | — (axiom) | [A] | carrier 3+2 | wrong family/charge lattice |
| $D_5$ | $C^+$ even-Hamming = half-spinor 16 | P2 | [I] | dim $S^+$=16, $Y$ | group/matter mismatch |
| $A_3$ | $\mathbb{P}^1 \setminus \mu_4$ four-puncture geom. | P1 | [I] | $N_{\text{fam}}$=3, $\mathbb{Z}_6$ | wrong multiplicity |
| $E_8$ | $\mu_4$ glue: disc=$\mathbb{Z}_4$, $q$-sum= 2 | $D_5, A_3$ | [I] | 240, 248 | not even-unimodular |
| $R, L$ | residue + winding matrix, anchor (1, 1, 2) | $A_3$ | [I] | det 8, minors (2, 3, 5) | wrong $D_6$ branch |
| $\alpha^{-1}$ | root of $F_{U(1)}(\alpha)$=0 | $c_3$, $M$=41 | [N] | 137.0359992 | no/second root (excl.) |
| masses | $\varphi_0^{\text{ret}}$-ladder $\lambda_Y^L \Lambda$ | $R, L$, $c_3$ | [I]/[P] | 9 masses, mixings | hierarchy mismatch |
| $\theta_{12}$ | TBM + seam $\varepsilon$=$c_3$ | $R, L$ | [N]/[P] | 0.307 | seam-misalign. lemma fails |
| $\Lambda, A_s$ | seam det. / $R^2$ scalaron | $c_3$ | [P] | $\Lambda$, $A_s, n_s, r$ | ambient RP fails |

*Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.*

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

## The compact status formula (three honest layers)

$$\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the $R$-bridge amplitude normalisation $U_{\text{point}}$ *reduces to* $v_{\text{geo}}$ (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so $G_{\text{metric}}$ is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale $v_{\text{geo}}$ — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces ($F_{\text{frontier}}$).

**Two points that are *not* in the residual (closed, but easy to mis-list as open).** *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([I]/[L] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [P]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the $E_8$ closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical $\Lambda$ branch ($(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* $\Lambda$, so $G_N$ is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([I]/[P], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just $\pi$, the discrete $E_8$-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale $v_{\text{geo}}$ — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two [$A$] anchors are *one* (dimensional analysis; no metre from pure mathematics).

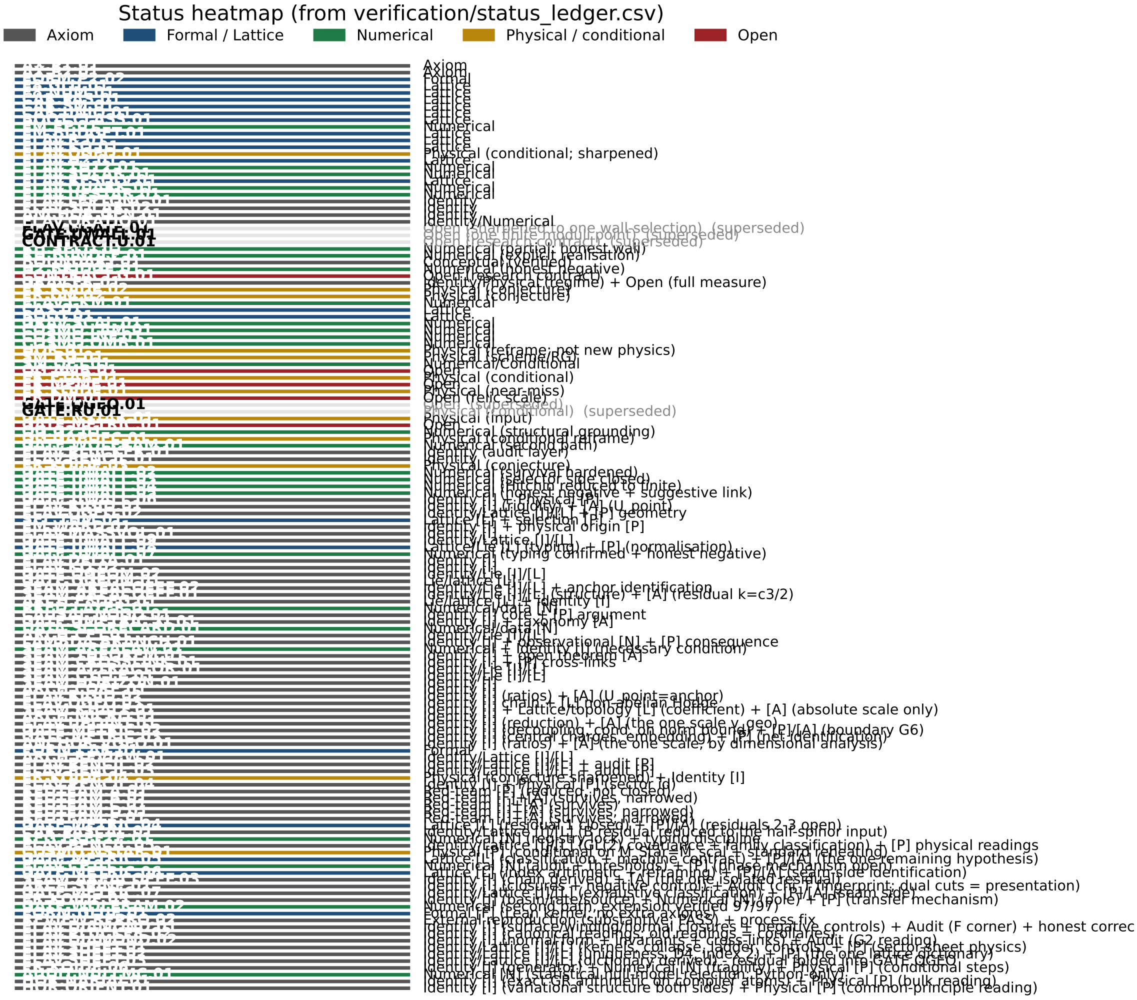



Figure 1: Status heatmap, generated directly from the single source of truth `verification/status_ledger.csv`. Grey = axiom, blue = formal/lattice, green = numerical, gold = physical/conditional, red = open. Each row carries a claim ID (`E8.GLU.01`, . . . ) that resolves to a ledger row with its dependencies and verification script. The ledger is *versioned*: each row has `active` and `canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive and drawn dimmed — the canonical status is the only one read as current.

**The hard reduction: everything open is two research gates plus typed interfaces**

After the compiler closure, the entire residual is

$$\boxed{\text{Rest} = (U_{\text{wall}}) \;\oplus\; (G_{\text{metric}}) \;\oplus\; (F_{\text{frontier}})}$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4* $\mu_4$ *simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank = $c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing $G_{\rm net}$ also closes the carrier selection [L]; $(F_{\rm frontier}) \to F_{\rm transfer}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, $\eta_B$ Boltzmann, axion relic, $m_p/m_e$), never compiler powers. The status card:

| Item | Status | Reading |
|---|---|---|
| parameter-freeness (self-consistency) | [I]/[L] | gapped transport ⇒ *unique* Perron–Frobenius attractor (v56); bootstrap web $g_{\rm car}$=5 forced (v6/v14) |
| Newton constant $G_N$ | [I]/[P] | not *independent*: branch pins $\bar{M}_{\rm Pl}$ rel. to $\Lambda$ (v60); readout after branch + one dim. anchor |
| ratio $c_u/c_d = \frac{55}{117}$ | [I] | $g_{\rm car}\lVert \mathrm{Pl}_{\mathbf{1},a}(K)\rVert_1/(N_{\rm fam}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve |
| $(U)$ flavor gate | [A]→ $v_{\rm geo}$ | **Gate 1 closed**: $U_{\rm point}\to v_{\rm geo}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$ |
| $R+R^2$ gravity | [I]/[P] | covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2) |
| physical QG sector | [F]/[P] | gap-decoupled admissible measure ($\Delta_{\rm eff}$=1.648>0) |
| ambient QG measure | [A](reduced) | **Gate 2 tier B**: holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. open ($G_{\rm metric}$) |
| Koide | [P] | $Q_{\rm src} + \varphi_0/24$ as source→pole transfer conjecture |
| axion DM | [P]/[A] | $f_a = M_{\rm scal}/128$ as a scenario |
| $\eta_B$ | [P] | readout closed; leptogenesis Boltzmann solve missing |
| $m_p/m_e$ | [A] | cross-sector QCD/EW ratio |
| $N_\star$ | [P] | reheating input, $N_\star \simeq 57$ (continuous) |
| $a = (1,1,2)$ | [L]/[P] | forced by $4 = 2+1+1$, conditional on the carrier |
| carrier selection $g$=5 (QBL) | [I]/[L] + [P] | theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = $c$); residue = the $G_{\rm net}$ premise itself — gate closes ⇒ carrier [L] |
| $E_8$ Lean | [F]→[L] | hardening, not a blocker (script-certified) |

*Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.*

Only *two* research gates remain — $(U_{\rm wall})$ and $G_{\rm metric}$; the frontier items are typed interfaces, not structural holes. Of these, $(U_{\rm wall})$ is the only genuine *physics* gate left: $G_{\rm metric}$ is now heat-kernel grounded (G2: the spectral action gives $R+R^2$ structurally) and *gap-decoupled* (G5: $\Delta_{\rm eff}$=1.648>0 isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority $(U_{\rm wall})$ first. [`verification/v36_spectral_action_g2.py`]

## 4 Before / after: what concretely changes

| Aspect | State of Papers 1–7 | After the compiler closure |
|---|---|---|
| $E_8$ | *peripheral*: only an $11D\to 4D$ reduction aside and the nilpotent-orbit *cascade* (scale ordering) | **central, constructed**: $C^+ = D_5$ spinor, $\mathbb{P}^1\backslash\mu_4 = A_3$, $\mu_4$ glue $\Rightarrow E_8$ **[L]** |
| $E_8$ numbers | $240, 248$ read off tables when needed | $240 = 16\cdot 5\cdot 3$, $248 = 240+8$ *derived* as carrier traces **[I]** |
| fine structure $\alpha^{-1}$ | *already* a parameter-free cubic fixed point $\alpha^3 - 2c_3\alpha^2 - 8bc_6 \ln(\cdot) = 0$, $\alpha^{-1} \approx 137.0365$ | *same* equation, now **existence & uniqueness proved** and refined to 137.0359992; dev. $2.9\times 10^{-10}$ vs CODATA-2022 ($1.9\sigma$) **[I]**/**[N]** |
| flavor residues | partly audit-flagged | anchor $\{1,1,2\}$ & matrix forced by 3 hypotheses **[I]**/**[P]** |
| many sectors | side by side, each its own closure | *one* compiler $\mathbb{Z}_{30} = 2\cdot 3\cdot 5$ generates them **[I]** |
| inputs | $c_3, g_{\mathrm{car}}$ (+ implicit structure) | exactly 2 axioms (P1, P2), clearly declared **[A]** |
| $A_s$, $n_s$, $r$ (inflation) | scale present | scalaron mass fixed by $c_3^7$ ($3.06\times 10^{13}$ GeV); $A_s, n_s, r$ become *predictions* **[P]** |
| $\theta_{12}$ (solar) | open (leading $\frac{1}{3}$) | **cond. derived** $\frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2} = 0.307$, given seam $\varepsilon = c_3$ **[N]**/**[P]** |

*Marker key: **[I]** exact identity · **[L]** Lie/lattice theorem · **[F]** formalised · **[N]** numerical fixed point · **[P]** physical/conditional · **[A]** axiom / open.*

**Reduction in one line**

The number of *structural* assumptions drops from "a fixed-point seed $c_3$, a carrier, the orbit cascade, plus a per-sector calibration in each block" to **two axioms**. Everything else ($N_{\mathrm{fam}}, \Omega_{\mathrm{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the $E_8$ glue, the scale grammar) becomes a *consequence*. *And even those two tighten:* the bootstrap note shows $g_{\mathrm{car}} = 5$ and the integer 8 in $c_3$ are *overdetermined $E_8$-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: $\pi$ from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.)

## 5 Paper by paper: what each original paper did, and how it works now

The seven original papers (plus the orientation note) total roughly **247 pages**. The table below puts each one next to its compiler-era replacement: *what* it treated and computed, *how many pages* it took, *where* that now lives and how much it compresses, and whether the *accuracy* of the result improves on top. The honest summary up front: in most sectors the *numbers are the same* — what shrinks dramatically is the *derivation length* and the number of independent assumptions; in three places (the solar angle, the inflation amplitude, the scalaron mass) there is a *genuinely new or sharper* result.

| P | Treated & computed | Old | How it works now (note; compression) | Acc. |
|---|---|---|---|---|
| 0 | orientation: architecture, decoder core $Y, [u_\Sigma], u$, theorem stack | 20 p | *one* compiler diagram: $\{c_3, g_{\rm car}\} \to D_5 \oplus A_3 \to E_8$ (intro + action_grammar) | = |
| 1 | boundary kernel: $c_3 = \frac{1}{8\pi}$, Calderón polarization, carrier decoder $6Y^2 - Y - \mathbf{1} = 0$ | 20 p | the two axioms P1 ($c_3$) + P2 ($g_{\rm car}=5$); carrier = 3+2 atom (compiler) | = |
| 2 | carrier/SM packet: $S^+=16$, $G_{\rm phys}$, $N_{\rm fam}=3$, $\Omega_{\rm adm}=48$, hypercharge, $b_1$ | 48 p | compiler reads 16=1+5+10, $N_{\rm fam}$, $\Omega_{\rm adm}$, $b_1 = \frac{41}{10}$ as carrier traces; $C^+ = D_5$ spinor (∼5 p) | = |
| 3 | EM closure: $\alpha^{-1}$ via $\sum L + N_\Phi = 41$; flavor transport, CKM/PMNS, $\theta_{13}$ | 38 p | $\alpha^{-1}$ as unique root of $F_{U(1)}(\alpha)=0$; flavor from transport theorem (parabolic + sm) | ↑ |
| 4 | admissibility/strong CP/QFT: $\theta_{\rm eff}=0$, OS reconstruction, mass gap | 40 p | explicit gap $6\log\frac{3}{2} \Rightarrow$ Dobrushin/OS; strong CP from structure+RP (closing_comp.) | = |
| 5 | geometric Hodge/metrology: $M_{\rm Pl}$, $v_{\rm geo}$, $\Lambda$, spectral Einstein | 30 p | Hodge/$R^2$ branch; $\Lambda$ *typed* (one branch label); boundary-normalised (closing_comp.) | = |
| 6 | cosmology interface: FRW, $\Lambda_{\rm IR}$, scalaron, reheating, leptogenesis, axion | 15 p | Starobinsky $M = c_3^{7/2}\bar{M}_{\rm Pl} = 3.06 \times 10^{13}$ GeV; $A_s$ *parameter-free* (closing_comp.) | ↑new |
| 7 | CMB operational closure: $C_\ell$, $a_{\ell m}$, seam mixer, Planck pipeline | 36 p | $n_s = 0.964$, $r = 0.0040$ from the $R^2$ attractor, no free amplitude (closing_comp.) | ↑new |

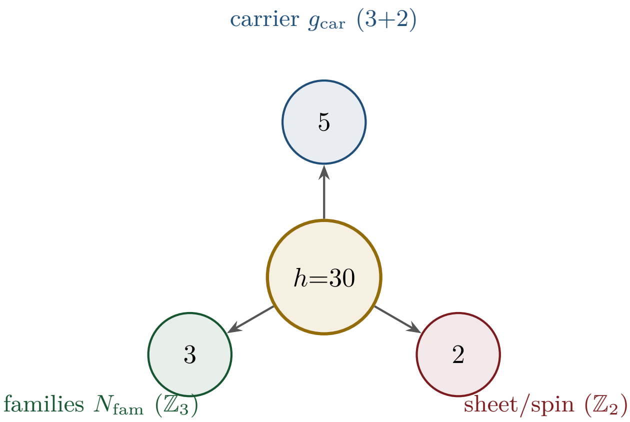
**Three places where accuracy genuinely improves "on top"**

1. **Solar angle $\theta_{12}$.** Paper 3 left it as the only open SM number (leading $\frac{1}{3}$). It is now *conditionally derived* (modulo the seam-misalignment lemma): TBM gives $\frac{1}{3}$, the charged-lepton 1–2 misalignment is the seam $\varepsilon = \frac{N_{\rm fam}}{|\mu_4|}\varphi_0^{\rm ret} = \frac{3}{4}\varphi_0^{\rm ret}$ (leading term exactly $c_3 = \frac{1}{8\pi}$), and TBM geometry gives $\sin^2\theta_{12} = \frac{1}{3} - \frac{2}{3}\varepsilon = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.3067$ — 0.1% from NuFIT.
2. **Inflation amplitude $A_s$.** Generic Starobinsky fits the scalaron mass to $A_s$. TFPT fixes it by the seam, $(M/\bar{M}_{\rm Pl})^2 = c_3^7$, so $A_s = \frac{N_\star^2}{24\pi^2}c_3^7 = 2.0 \times 10^{-9}$ is a *prediction* (Planck $2.1 \times 10^{-9}$), and the measured $A_s$ even *predicts* $N_\star = 56$.
3. **Scalaron mass.** $M = c_3^{7/2}\bar{M}_{\rm Pl} = 3.06 \times 10^{13}$ GeV comes out *exactly* at the canonical Starobinsky value — a number that was an input before is now an output.

**What "simpler" means concretely.** The compression is not cosmetic. Paper 2 spent 48 pages establishing the carrier/SM packet by rigidity arguments; the compiler reads the same 16=1+5+10, $N_{\rm fam}=3$, $\Omega_{\rm adm}=48$, $b_1 = \frac{41}{10}$ off the Pascal row of a 5-slot carrier in a page. Paper 3's flavor transport (38 p) becomes one transport theorem plus a fixed residue matrix whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers. The "many independent closures" of Papers 1–7 become *one* generator with a handful of corollaries.

## 6 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of $E_8$ is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

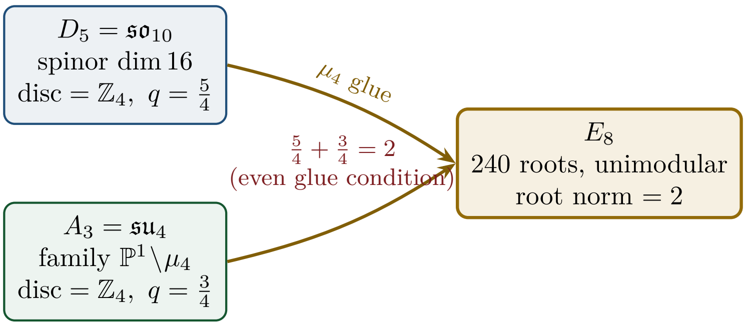



From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the $E_8$ root count. [I]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [I]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial $\Phi_{30}$; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \setminus \mu_4$ is the $A_3$ Coxeter element. [I]

## 7 The $\mu_4$ glue: how $E_8$ is really built

The decisive, now *proved* step: $D_5$ and $A_3$ have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the $E_8$ root norm.




**Glue theorem (proved)**

$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the $E_8$ root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind "positing of 248". [L]

## 8 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\rm car}\}$? The following map summarises what is now a consequence rather than an assumption.

| Sector | How it now arises | Status |
|---|---|---|
| gauge group / families | carrier 3+2, $\dim S^{+}=16$, $N_{\mathrm{fam}}=3$, $\Omega_{\mathrm{adm}}=48$ | [I] |
| hypercharge | roots of the charge polynomial $bsY^2+(s-b)Y-1$ | [I] |
| $U(1)$ index $b_1$ | $b_1=\dfrac{g_{\mathrm{car}}\,2^{g_{\mathrm{car}}-2}+1}{10}=\dfrac{41}{10}$ | [I] |
| fine structure $\alpha^{-1}$ | unique root of $F_{U(1)}(\alpha)=0$ | [I]/[N] |
| flavor matrix | transport theorem (H1 proved, H2/H3 force the matrix) | [I]/[P] |
| electroweak scale | $v_{\mathrm{EW}}\sim e^{-\alpha^{-1}/5}$ (carrier divides $\alpha^{-1}$ by 5) | [I] |
| cosmological constant | $\Lambda\sim e^{-2\alpha^{-1}}$ (density quotient; see typing note) | [I]/[P] |
| Hubble scale | $H_0\sim\sqrt{\Lambda}$ | [P] |
| inflation amplitude | $A_s=\dfrac{N_\star^2}{24\pi^2}c_3^7$ in the $R^2$ branch | [P] |
| $E_8$ | $D_5\oplus A_3+\mu_4$-glue | [L] |

*Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.*

**Notable:** the scale grammar is *one* exponential engine. The same number $\alpha^{-1}\approx 137$ generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

**Typing note on $\Lambda$ (a deliberate danger zone).** "$\Lambda$" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [I] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4$ (which carries a prefactor *branch*, hence [P]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

**Cosmology is the Pascal action grammar of the carrier — not a separate module**

With $x=v_{\mathrm{geo}}/(5\beta_{\mathrm{rad}}^2\bar{M}_{\mathrm{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph $K_5$:

$$\boxed{\frac{H_0}{\bar{M}_{\mathrm{Pl}}}=x^5R_H,\qquad \frac{\rho_\Lambda}{\bar{M}_{\mathrm{Pl}}^4}=x^{10}R_\Lambda},\qquad (1,5,10)=\tbinom{5}{0},\tbinom{5}{1},\tbinom{5}{2}.$$

EW is the unit $x^1$, Hubble is the product over the $\binom{5}{1}=5$ vertices, $\Lambda$ is the product over the $\binom{5}{2}=10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the $E_8$ root count $16\cdot 5\cdot 3=240$. "Hubble = slot product, $\Lambda$ = pair product." [I]

**The entire fermion spectrum in one formula**

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j}=\frac{v_{\mathrm{geo}}}{\sqrt{2}}\,\lambda_Y^{L_{f,j}}\,\Lambda_{f,j},\qquad \lambda_Y=\sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})},\qquad \varphi_0^{\mathrm{ret}}=\tfrac{1}{6\pi}+\tfrac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3-9t^2+10t-8$ carries only compiler numbers ($\mathrm{tr}=9=N_{\mathrm{fam}}^2$, $\det=8=h(D_5)$, principal 2-minors 2,3,5 with product $30=h(E_8)$). Through the $\varphi_0^{\mathrm{ret}}$-ladder $\hat{m}=\frac{v_{\mathrm{geo}}\pi}{\sqrt{2}}c\,(\varphi_0^{\mathrm{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the ($U_{\mathrm{wall}}$) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer ($m_H$ rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2}$; see below). *(Full derivation: document 2,* `tfpt_2_standard_model`*.)*

## 9 Physical, geometric, topological: which solution is more plausible?

| Level | $E_8$ peripheral (v4.5) | $E_8$ compiled (new) |
|---|---|---|
| physical | $\alpha^{-1}$ already a cubic fixed point, but each sector closed on its own | $\alpha^{-1}$ as the fixed point of *one* equation; one generator for many scales |
| geometric | Lie algebra external; $E_8$ only orders the cascade | the $C^+$ spinor + puncture geometry $\mathbb{P}^1 \setminus \mu_4$ give $D_5, A_3$, glued to $E_8$ |
| topological | — | the common discriminant $\mathbb{Z}_4$ ($\mu_4$ winding) forces the glue; $\Phi_{30}$/Coxeter drives the sectors |
| parsimony | many structures | 2 axioms, the rest a consequence |

**Why the compiled solution is more plausible**

It is (i) *more parsimonious* (fewer independent assumptions $\Rightarrow$ less overfitting risk), (ii) *more rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *more checkable* (machine-checkable $E_8$ appendix, explicit fixed-point equation for $\alpha^{-1}$). More explained from less — exactly the criterion that makes a theory more probable.

## 10 What role $E_8$ now plays exactly

$E_8$ is no longer the starting point but the *output format*:

- **Bookkeeping:** $E_8$ is the unimodular container in which carrier ($D_5$) and family ($A_3$) fit together consistently. Its numbers $240, 248$ are carrier traces, not inputs.
- **Consistency test:** unimodularity (det 1) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. $E_8$ thus tests the internal coherence of the two atoms.
- **Not an end in itself:** $E_8$ does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in $E_8$.

**New — the secondary atlas.** Beyond that, $E_8$ is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit:

**$E_6 \times A_2$ reads the flavor matrix**

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\mathrm{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one $E_8$ projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

## 11 What this means for the theory-of-everything question

**The state of the five questions — closed, reduced or typed**

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector $R$, the $Q$ spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from $D_4$, v69/v70), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall ($U_{\text{point}}$, an anchor; Research Contract 1). [I]/[A]
2. **Gravity — exact in the $R^2$ regime.** Spectral action $\to R+R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\to$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n\geq 3} \to R^2$, valid for curvature $\ll M^2$). [I] in the regime
3. **Strong CP — decoupled.** $\theta_{\text{eff}}=0$ follows from three structural facts ($\gamma_5$-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [I] (given RP)
4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap $6\log\frac{3}{2}$ of the $C_6$ transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [I] (transfer model) / [P] (continuum)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`1886 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [I]/[A]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.)*

## The TOE-completion ledger — what is done, and what genuinely remains

A theory-of-everything must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

| **TOE piece** | **Status** | **If not complete: what closes it** |
|---|---|---|
| gauge group, $N_{\text{fam}}$, hypercharge, $b_1$ | [I] | — complete |
| $E_8 = (D_5 \oplus A_3) + \mu_4$, group closure | [L] | — complete |
| bootstrap $(g, \mu) = (5, 4)$, "8" in $c_3$ | [L] | — complete (reverse glue $\mu^2 - 5\mu + 4 = 0$) |
| fine structure $\alpha^{-1}$ | [N] | — **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin [P] |
| fermion masses ($\varphi_0^{\text{ret}}$-ladder) | [I]/[A] | leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor |
| mixings $\theta_{13}, \theta_{23}, \theta_{12}$ | [N]/[P] | $\theta_{12}$ cond. derived; $\theta_{23}$ maximal |
| flavour matrix / H2 | [I] | up to Mehta–Seshadri unitarity (U), a Paper-3 input |
| inflation $A_s, n_s, r$, scalaron $M$ | [N] | parameter-free given $M_{\text{Star}} = M_{\text{scal}}$ |
| *genuinely open — the actual TOE remainder* | | |
| dim. EW/QCD masses | [P] | standard RG threshold matching (no new physics) |
| QFT measure (admissible sector) | [I] | **closed: RP from cusp spectrum, OS reconstruction** |
| ambient QFT + full gravity (metric sector) | [A] | *not constructed here* (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure |
| analytic boundary origin of $\pi$ ($c_3 = \frac{1}{8\pi}$) | [P] | **hardened:** $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8" = rank $E_8$; $\pi$ stays the one primitive |
| $\eta_B$, $m_p/m_e$, Koide, dark matter | [A] | frontier handles only (see document 4) |

*Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.*

**The honest bottom line:** the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one $\varphi_0^{\text{ret}}$-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays [A] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*: $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K\, dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank $E_8$ and only the Gauss–Bonnet $\pi$ left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and $\pi$ is the lone continuous primitive.

## 12 Does this make the theory more probable?

### Plausibility balance

**Pro:** (a) two inputs instead of many; (b) $\alpha^{-1}$ to $2.9 \times 10^{-10}$ (CODATA-2022, $1.9\sigma$) as a *fixed point*, not a fit; (c) $E_8$ constructed, not postulated; (d) one compiler generates several sectors $\Rightarrow$ fewer degrees of freedom, more predictive power.

**Contra/open:** what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

## 13 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

| **Observable** | **TFPT value & derivation** | **Experiment (running / upcoming)** | **St.** |
|---|---|---|---|
| solar angle $\sin^2\theta_{12}$ | $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.3067$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret} \approx c_3$ (cond.) | **JUNO** (taking data since Aug 2025; sub-percent $\theta_{12}$) — the sharpest live test | [N]/[P] |
| reactor angle $\sin^2\theta_{13}$ | $\varphi_0^{\rm ret} e^{-5/6} = 0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma{=}\frac{5}{6}$ | Daya Bay / RENO / JUNO (PDG 0.0220) | [I] |
| atmospheric $\sin^2\theta_{23}$ | $\approx \frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected) | NOvA / T2K / **DUNE** (octant ambiguity, NuFIT 6.0) | [P] |
| fine structure $\alpha^{-1}$ | 137.0359992 as the unique root of $F_{U(1)}(\alpha){=}0$ (no fit, no second branch) | CODATA-2022 137.035999177(21): dev. $2.9 \times 10^{-10}$ from the recommended adjustment ($1.9\sigma$ of its uncertainty) | [I]/[N] |
| tensor ratio $r$ | $r = \frac{12}{N_\star^2} = 0.0040$ ($N_\star{=}55$); $R^2$ attractor, $M{=}c_3^{7/2}\bar{M}_{\rm Pl}$ | now $r{<}0.036$ (BICEP/Keck BK18); **CMB-S4** ($\sigma_r{\le}5{\times}10^{-4}$, obs. $\sim$2033) | [P] |
| tilt $n_s$ | $n_s = 1 - \frac{2}{N_\star} = 0.964$ (same attractor) | Planck (0.965); CMB-S4 sharpens | [P] |
| amplitude $A_s$ | $\frac{N_\star^2}{24\pi^2} c_3^7 = 2.0{\times}10^{-9}$, parameter-free | Planck ($2.1 \times 10^{-9}$) | [P] |
| neutron EDM | $\theta_{\rm eff} = 0 \Rightarrow d_n$ essentially null (structure + RP) | **PSI nEDM** / SNS (running) | [I] |
| $0\nu\beta\beta$ / ordering | Majorana branch; normal-ordering preference | **LEGEND / nEXO**, DUNE, KATRIN | [P] |
| flavor invariants | $\det R{=}h(D_5){=}8$, principal minors $(2,3,5)$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$ (exact) | any future CKM/PMNS global fit must satisfy these | [I] |
| $H_0$ vs. $\Lambda$ | $v_{\rm EW} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine | SH0ES / **DESI** / Planck (Hubble tension) | [P] |

*Marker key: **[I]** exact identity · **[L]** Lie/lattice theorem · **[F]** formalised · **[N]** numerical fixed point · **[P]** physical/conditional · **[A]** axiom / open.*

**The two decisive near-term tests**

**(1) JUNO and $\theta_{12}$.** JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the conditional $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.3067$ (seam $\varepsilon = c_3$). A measured central value away from $\sim 0.307$ at high significance falsifies the seam mechanism — a *live* test.

**(2) $r$.** The $R^2$ scalaron with $M$ fixed by $c_3^7$ predicts $r = 0.0040$ ($N_\star{=}55$), already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \le 5 \times 10^{-4}$, first observations $\sim$2033–34). A detection of $r \gtrsim 0.01$ would be incompatible with the $R^2$ branch on which $M_{\rm Pl}$ and $A_s$ rest.

**Structural (non-experimental) checks.** Independent of any apparatus, the machine-checkable $E_8$ appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two

TFPT atoms really generate $E_8$; and the discipline rule "*every load-bearing number appears in at least one $E_8$ projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge itself is additionally *quantified* by an explicit null model: exhaustively enumerating a declared formula grammar (provably containing every scored TFPT formula) against conservative data windows gives a joint look-elsewhere-corrected null probability $\leq 10^{-30.7}$ that a random theory of equal formula complexity reproduces the scorecard — a statistical statement conditional on the declared grammar, not a claim of certainty [N] [verification/v100_numerology_null_mc.py].

**Freeze file: committed kill criteria**

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the $R^2$ branch carrying $M_{\text{Pl}}, A_s$;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

**Blind-prediction registry (frozen 2026-06-09, machine-enforced).** Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84_frozen_registry.py] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* $\theta_{12}$ prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. $r$ and $n_s$ enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

**What likely follows next**

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor ($U_{\text{point}}$) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over $S^+$;
- a sharper geometry↔gravity bridge once the Hodge/$R^2$ branch is lifted from [P] to [I].

## Conclusion

The state of Papers 1–7 was an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the $E_8$ container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

## Appendix: computational verification (reproducibility)

Every claim in this document marked [I] (exact identity), [L] (Lie/lattice theorem) or [N] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\mathrm{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map.

| Script | What it machine-checks |
|---|---|
| `v1_e8_glue.py` | glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$ |
| `v2_carrier_pascal.py` | $g_{\rm car}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\rm fam}{=}3$, rank $E_8{=}8$, $\Omega_{\rm adm}{=}48$, $b_1{=}\frac{41}{10}$ |
| `v3_em_alpha.py` | EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation |
| `v4_flavor_matrix.py` | residue matrix $R$: det $=8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$ |
| `v5_e8_cascade.py` | cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to$ 240, IR tail 46080, variance 6552 |
| `v6_bootstrap.py` | reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\rm car}{=}5$ three ways; $8{=}\,$rank $E_8{=}h(D_5){=}\varphi(30)$ |
| `v7_gravity_cosmo.py` | scalaron exponent 7, $M^2_{\rm scal}/\bar{M}^2_{\rm Pl}{=}c_3^7$, $A_s, n_s, r$, $\Omega_b$, $\sin^2\theta_{12}$ |
| `v8_horizon.py` | horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, $S_{dS}$, Page, $\beta_{\rm rad}{=}0.2424^\circ$ |
| `v9_neutrino_texture.py` | solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$, $\theta_{23}{=}45^\circ$, $\theta_{13}{=}0$ |
| `v10_projection_ involution.py` | $Q,\Sigma$ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; $\chi_Q, \chi_K$; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$ |
| `v11_unique_KQ.py` | $K, Q$ are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration) |
| `v12_mass_ generation_ polynomials.py` | sector/generation polynomials of $K$ and the anchor-block det ladder $(9,10,16,40)$ |
| `v13_open_gates.py` | gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\rm fam}$ |
| `v14_carrier_ uniqueness.py` | $g_{\rm car}{=}5$ unique ($N_{\rm fam}\in\mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\mathrm{Tr}\,Y{=}\,\mathrm{Tr}\,Y^3{=}0$ |
| `v15_bootstrap_ classification.py` | $D_5{\oplus}A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$ |
| `v16_solar_dual_ anchor.py` | $a^\top R^{-1}{=}a^\top L^{-1}{=}({-}\frac{1}{2},{-}\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$; full PMNS |
| `v17_hexagonal_ resolvent.py` | finite resolvent $D_y^{-1}{=}(\sum y^{5-m}\delta^m U_6^m)/(y^6{-}\delta^6)$ (quark $c$ backbone) |
| `v18_quark_yukawa.py` | quark source ratios $\frac{m_u}{m_d}{=}0.470085$, $\frac{m_c}{m_s}{=}13.61$, $\frac{m_t}{m_b}{=}40.80$ exact; lepton $c{=}(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy |
| `v19_monodromy_ moduli.py` | exact pole $z_\star{=}\frac{794-7\sqrt{9961}}{2187}$ $(3^7)$; $D_4$ $SU(3)$ monodromy constructible but $D_4$-fixed locus positive-dim $\Rightarrow$ exact quark $c$ reduce to $(U)$ |
| `v20_lepton_c_ derivation.py` | lepton $c$'s derived: $\delta{=}\frac{1}{2}$ resolvent $c{=}\lvert\mu_4\rvert^w/(\frac{5}{4}{-}\cos\frac{r\pi}{3}) \Rightarrow \frac{16}{7},\frac{4}{3}$; product $\frac{2^{g_{\rm car}}}{N^2_{\rm fam}}{=}\frac{32}{9} \Rightarrow \frac{7}{6}$ |
| `v21_solar_product_ quark.py` | solar coeff $= q(A_3){=}\frac{3}{4}$ (glue-norm, $q(D_5){+}q(A_3){=}2$), $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$; product $\frac{2^{g_{\rm car}}}{N^2_{\rm fam}}$; quark law fails ($\frac{1}{7}$ vs 0.724) |
| `v22_open_gates_ audit.py` | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual) |
| `v23_anchor_ generator.py` | anchor $a{=}(1,1,2)$: $e_k{=}(4,5,2)$, $c_3{=}\frac{1}{2e_1\pi}$; $p_n{=}2{+}2^n \Rightarrow 240, 248$, binary ladder; inputs $\to \{a,\pi\}$ |
| `v24_quark_ratio_ closure.py` | quark ratios $\frac{55}{117},\frac{34}{47},\frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational $c$ gauge, $\Lambda_q$ in $O(1)$ |
| `v25_frontier_ conjectures.py` | conjectures: Koide $Q_{\rm src}{+}\frac{\varphi_0^{\rm ret}}{24}{\approx}\frac{2}{3}$; axion $f_a{=}M_{\rm scal}/128{\to}m_a{\approx}23.8\,\mu$eV |
| `v26_ flavor_frontier_ unification.py` | the "11" is not uniquely forced ($\Rightarrow$ [P]); flavor frontier unifies to one $(U)$ gate; cusp config on the parabolic stability wall |
| `v27_wall_ representative.py` | explicit balanced wall rep $W_{\rm wall}$ (cols = perm$(0,1,2)$, rows $w{=}(2,1,1)$, pardeg 0); selector det $R{=}8$, Spec$(Q_+){=}\{1,2,3\}$ |
| `v28_gravity_fR.py` | $R{+}R^2$ closed: $f(R)$, $M{=}c_3^{7/2}\bar{M}_{\rm Pl}$ scalaron, $N_\star{=}57{\to}n_s, r, A_s$; full metric measure open; $248c_3^2{<}6\log\frac{3}{2}$ |
| `v29_ research_contract_` | research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick $W_{\rm wall}$) and $G5$ gap-dominance $2\lVert V\rVert{<}\Delta$ |

**How to run.** From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all six documents are written against). The **[F]** (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).